Akoizo Original Practice - Form 1

Division B | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) One mm over one km^2 equals…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. 1000 m^3

B. 1 m^3

C. 1 million m^3

D. 1000 liters

Answer: _____

2. (2 points) Two second-order streams create…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. First order

B. Third order

C. Second order

D. Fourth order

Answer: _____

3. (2 points) An oxbow commonly forms through…

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. Only coral growth

B. Only groundwater pumping

C. Meander cutoff

D. Only glacier freezing

Answer: _____

4. (2 points) High porosity always means high permeability?

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Always

B. Only for clay

C. Only below water table

D. No

Answer: _____

5. (2 points) Impervious connectivity can affect…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Runoff timing

B. Only mineral hardness

C. Only gene frequency

D. No water process

Answer: _____

6. (2 points) A budget requires…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Only a map color

B. Only a starting temperature

C. A boundary and interval

D. Only a river name

Answer: _____

7. (2 points) Groundwater divides always match surface divides?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Always

B. Only on maps

C. Only in winter

D. No

Answer: _____

8. (2 points) Terrace elevation alone establishes exact age?

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. No

B. Always

C. Only for sandstone

D. Only below sea level

Answer: _____

9. (2 points) Darcy flux equals…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Porosity alone

B. Q/A

C. Q x A

D. K/A

Answer: _____

10. (2 points) Load rate needs…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only map area

B. Only color

C. Concentration and discharge

D. Only pH

Answer: _____

11. (2 points) Infiltrated water can later…

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

A. Support streamflow

B. Never move again

C. Only evaporate instantly

D. Create salt by definition

Answer: _____

12. (2 points) Radial drainage can suggest…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. No topography

B. An elevated central feature

C. Only flat sediment

D. Only a straight fault

Answer: _____

13. (2 points) Migration evidence is strongest from…

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. One water sample

B. Only river name

C. Comparable positions over time

D. One unlabeled image

Answer: _____

14. (2 points) Flow generally follows…

Unit: Groundwater, aquifers, karst, and Darcy flow.
Worked-case data: K=0.001 m/s, A=10 m^2, head drop=2 m over 100 m. Find Q.

A. Only surface slope

B. Increasing head always

C. Only north

D. Decreasing hydraulic head

Answer: _____

15. (2 points) Before/after change proves causality alone?

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. No

B. Always

C. Only after one storm

D. Only in a reservoir

Answer: _____

16. (4 points) What is missing from a simple P-Q budget?

Unit: Water movement and water budgets.
Worked-case data: Rainfall is 20 mm over 2 km^2 with runoff fraction 0.30. Find direct runoff.

Answer: _____
_____
_____

17. (4 points) Why specify the outlet?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

Answer: _____
_____
_____

18. (4 points) What can form a point bar?

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

Answer: _____
_____
_____